1 Forward kinematics models

1.1 DH description

The six revolute joints are described using the *standard* DH convention. Frame $\{0\}$ coincides with `base_link`, and $q_i = 0$ denotes the URDF home configuration. The endpoint E is `link_6`. For each step i, the transform from frame $\{i-1\}$ to frame $\{i\}$ is

$$^{i-1}T_i = R_z(\theta_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i), \qquad \theta_i = q_i + \theta_i^0. \tag{1}$$

Joint i rotates about z_{i-1}. For $i < 6$, the new axis z_i lies on joint $i+1$; the terminal axis z_6 is chosen parallel to z_5. DH frame numbers therefore differ from physical joint numbers.

How to read one step. Choose x_i along a common normal to z_{i-1} and z_i. Its intersection with z_i defines O_i. If two distinct axes intersect, their common-normal length is zero and x_i is perpendicular to both axes through their intersection. For coincident axes, the origin and transverse x axis are free; here the last origin is the endpoint. Set $y_i = z_i \times x_i$ to complete a right-handed frame. The four parameters then have the following geometric meaning:

Parameter	Measured about / along	Construction from $\{i-1\}$ to $\{i\}$
θ_i	about z_{i-1}	Turn x_{i-1} towards x_i.
d_i	along z_{i-1}	Move from O_{i-1} to the foot of the common normal.
a_i	along x_i	Follow the common normal to O_i.
α_i	about x_i	Turn z_{i-1} towards z_i.

Translations are signed along the indicated positive axes; rotations follow the right-hand rule. Figure 1 shows the construction at $q = 0$.

Table 1: Six DH steps. Lengths are in metres; angular constants are in radians.

i	d_i	θ_i	α_i	a_i	Geometric construction at home
1	1	q_1	$-\pi/3$	0.8	Rise $0.5 + 0.5$ from the base, then move 0.8 along x_1 to joint 2. Its axis gives a $-60°$ twist.
2	1.25	q_2	$+\pi/2$	2	Move 1.25 along joint 2's axis and 2 along x_2 to joint 3; twist by $+90°$.
3	0.75	$q_3 + \pi/2$	$+\pi/2$	0	Extend joints 3 and 4's axes to their intersection O_3. Turn x_2 by $+90°$ to select x_3; no common-normal translation is needed.
4	2.5	q_4	$-\pi/2$	0	From O_3, follow joint 4's axis by $1.5 + 1$ to the wrist centre O_4; twist by $-90°$.
5	0	q_5	$+\pi/2$	0	Joints 5 and 6's axes meet at the wrist centre. Keep $O_5 = O_4$ and twist by $+90°$.
6	1.5	q_6	0	0	Follow joint 6's axis from the wrist centre to the endpoint. Choose $z_6 = z_5$ and no fixed twist.

Frame origins versus joint centres. A DH origin lies on the selected joint axis but need not coincide with a URDF joint origin. In particular, O_3 lies 1.5 m before the

physical centre of joint 4. This distance enters d_4, giving $d_4 = 1.5 + 1 = 2.5$ m, while $a_3 = 0$. Similarly, $d_5 = 0$ even though the centres of joints 5 and 6 are 1.5 m apart: the two axes already intersect at $O_4 = O_5$.

Endpoint alignment. Frame $\{6\}$ and the URDF endpoint have the same origin but different axes: $x_E = z_6$, $y_E = x_6$ and $z_E = y_6$. Hence the forward kinematics uses the fixed alignment on the right:

$$ {}^0T_E = \left(\prod_{i=1}^{6} {}^{i-1}T_i\right) {}^6T_E, \qquad {}^6T_E = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{2} $$

1.2 Product of exponentials model

The product of exponentials (PoE) describes the robot through its joint axes at the home configuration. All vectors below are expressed in `base_link`, with $q = 0$ and endpoint `link_6` as in the DH model.

1. Describe each revolute joint by a screw axis. Let ω_i be the unit direction of joint i and let r_i be a point on its axis at home. With unit joint speed, a point p has velocity $\omega_i \times (p - r_i) = \omega_i \times p + v_i$. This gives the six-vector ξ_i and its matrix form:

$$ v_i = -\omega_i \times r_i, \qquad \xi_i = \begin{pmatrix} \omega_i \\ v_i \end{pmatrix}, \qquad \widehat{\xi}_i = \begin{pmatrix} [\omega_i]_\times & v_i \\ 0^T & 0 \end{pmatrix}. \tag{3} $$

Here $[\omega]_\times$ represents the cross product:

$$ [\omega]_\times = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}, \qquad [\omega]_\times p = \omega \times p. \tag{4} $$

Any point on the same axis gives the same v_i: adding a multiple of ω_i to r_i does not change the cross product.

2. Find the six axes at home. The fixed tilt $\gamma = \pi/6$ maps the local y and z axes to $(0, \sqrt{3}/2, 1/2)^T$ and $(0, -1/2, \sqrt{3}/2)^T$. The points r_i follow by adding the URDF joint offsets in the base frame. For example, $r_3 = r_2 + (2, 0, 0)^T + (5/4)\omega_2$. Applying (3) then gives Table 2.

At home, axes 4 and 6 lie on the same line and point in the same direction. Their chosen origins differ only along that line, so $\xi_4 = \xi_6$. This equality describes the home configuration; the axes need not remain aligned when joint 5 moves.

3. Obtain the motion of one joint. For a unit axis ω_i, Rodrigues' formula gives the rotation $R_i(q_i)$. The corresponding rigid motion is the matrix exponential

$$ \begin{aligned} R_i(q_i) &= I_3 + \sin q_i [\omega_i]_\times + (1 - \cos q_i)[\omega_i]_\times^2, \\ e^{\widehat{\xi}_i q_i} &= \begin{pmatrix} R_i(q_i) & (I_3 - R_i(q_i))r_i \\ 0^T & 1 \end{pmatrix}. \end{aligned} \tag{5} $$

Table 2: Home screw data. Each vector is written as a row; r_i uses the physical joint origin. Lengths are in metres.

i	ω_i^T	r_i^T	v_i^T
1	$(0, 0, 1)$	$(0, 0, \frac{1}{2})$	$(0, 0, 0)$
2	$(0, \frac{\sqrt{3}}{2}, \frac{1}{2})$	$(\frac{4}{5}, 0, 1)$	$(-\frac{\sqrt{3}}{2}, -\frac{2}{5}, \frac{2\sqrt{3}}{5})$
3	$(0, -\frac{1}{2}, \frac{\sqrt{3}}{2})$	$(\frac{14}{5}, \frac{5\sqrt{3}}{8}, \frac{13}{8})$	$(\frac{7}{4}, -\frac{7\sqrt{3}}{5}, -\frac{7}{5})$
4	$(1, 0, 0)$	$(\frac{43}{10}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(0, \frac{13+3\sqrt{3}}{8}, \frac{3-5\sqrt{3}}{8})$
5	$(0, -\frac{1}{2}, \frac{\sqrt{3}}{2})$	$(\frac{53}{10}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(\frac{7}{4}, -\frac{53\sqrt{3}}{20}, -\frac{53}{20})$
6	$(1, 0, 0)$	$(\frac{34}{5}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(0, \frac{13+3\sqrt{3}}{8}, \frac{3-5\sqrt{3}}{8})$

The translation term follows by writing the rotated point as $r_i + R_i(q_i)(p - r_i)$. It keeps every point on the joint axis fixed.

4. Set the home pose and combine the joint motions. At $q = 0$, the endpoint has orientation $R_x(\gamma)$ and position r_6. Its home pose is therefore

$$M = {}^0T_E(0) = \begin{pmatrix} 1 & 0 & 0 & \frac{34}{5} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} & \frac{5\sqrt{3}-3}{8} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} & \frac{13+3\sqrt{3}}{8} \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{6}$$

The complete forward kinematics is

$$\boxed{{}^0T_E(q) = e^{\widehat{\xi}_1 q_1} e^{\widehat{\xi}_2 q_2} \cdots e^{\widehat{\xi}_6 q_6} M.} \tag{7}$$

The factors are written in joint order, from 1 to 6. The screw axes are fixed at their home values; the product accounts for how earlier joints move the later axes. At $q = 0$, all exponentials are identity matrices and the expression returns M.

1.3 Verification

Using the data above, the PoE, DH and direct URDF models were compared at 514 configurations: home, each joint alone at $\pm\pi/2$, one mixed pose, and 500 random configurations in $[-\pi, \pi]^6$. Across all model pairs, the largest position residual was 5.11×10^{-11} m and the largest rotation-matrix Frobenius residual was 1.18×10^{-11}. The DH–PoE residuals were below 10^{-14}; the URDF stores the fixed tilt as a decimal approximation of $\pi/6$.

2 Analytical inverse kinematics

The axes of joints 4–6 intersect at the wrist centre, allowing position and orientation to be solved separately. We first determine the arm angles (q_1, q_2, q_3) from the desired wrist-centre position, then recover the wrist angles (q_4, q_5, q_6) from the desired orientation.

2.1 Wrist-centre positioning

Let R_d and p_d denote the desired orientation and position of `link_6` relative to `base_link`. Since the endpoint offset lies along the tool's x axis, the target wrist centre is

$$p_w = p_d - d_6 R_d e_x = (x_w, y_w, z_w)^T, \qquad e_x = (1, 0, 0)^T. \tag{8}$$

Throughout, $c_i = \cos q_i$ and $s_i = \sin q_i$.

Derivation

1. Relate the target wrist position to the three arm angles. For a candidate q_1, undo the base rotation, translate to the shoulder at $s = (a_1, 0, d_1)^T$, and undo the fixed tilt $\gamma = \pi/6$. The resulting auxiliary frame follows the base rotation and fixed tilt, but not the joint-2 rotation. With $h = z_w - d_1$ and $(c_\gamma, s_\gamma) = (\sqrt{3}/2, 1/2)$,

$$\begin{aligned} r(q_1) &:= R_x(-\gamma)\Big(R_z(-q_1)p_w - s\Big) = (X, Y, Z)^T, \\ X &= c_1 x_w + s_1 y_w - a_1, \\ Y &= c_\gamma(-s_1 x_w + c_1 y_w) + s_\gamma h, \\ Z &= -s_\gamma(-s_1 x_w + c_1 y_w) + c_\gamma h. \end{aligned} \tag{9}$$

These coordinates still depend on the unknown q_1.

In the moving URDF frame of `link_2`, the shoulder-to-elbow vector is $a = (a_2, d_2, 0)^T$. The elbow-to-wrist vector is $R_z(q_3)b_0$, where $b_0 = (d_4, 0, d_3)^T$ is expressed in the URDF frame of `link_3_d3`. Applying the shoulder rotation maps their sum into the auxiliary frame:

$$r(q_1) = R_y(q_2)\Big(a + R_z(q_3)b_0\Big) = R_y(q_2)\begin{pmatrix} a_2 + d_4 c_3 \\ d_2 + d_4 s_3 \\ d_3 \end{pmatrix}. \tag{10}$$

The rotation $R_z(q_3)$ above is expressed using the URDF link frames, with $q_3 = 0$ at the home configuration. The DH frames use a different reference orientation: at home, their angular parameter is already $\theta_3 = \pi/2$. Consequently, $\theta_3 = q_3 + \pi/2$, whereas the rotation in this geometric derivation uses q_3 directly.

2. Express $\sin q_3$ and $\cos q_3$ as functions of q_1. Rotation about y preserves both the y component and the norm. Equation (10) therefore gives two relations independent of q_2:

$$\begin{aligned} Y &= d_2 + d_4 s_3, \\ N := X^2 + Y^2 + Z^2 &= a_2^2 + d_2^2 + d_3^2 + d_4^2 + 2a_2 d_4 c_3 + 2d_2 d_4 s_3. \end{aligned} \tag{11}$$

Setting $K = d_2^2 - a_2^2 - d_3^2 - d_4^2$ yields

$$s_3 = \frac{Y - d_2}{d_4}, \qquad c_3 = \frac{N + K - 2d_2 Y}{2a_2 d_4}. \tag{12}$$

Here $a_2 = 2$ and $d_4 = 2.5$ are nonzero. For a trial value of q_1, these expressions give the required values of $\sin q_3$ and $\cos q_3$. They correspond to a real elbow angle only if $s_3^2 + c_3^2 = 1$. This consistency condition provides the equation used to determine q_1.

3. Determine the candidate base angles q_1. Since $R_x(-\gamma)$ preserves the norm, $N = \|r(q_1)\|^2 = \|R_z(-q_1)p_w - s\|^2$. Defining $N_0 = x_w^2 + y_w^2 + h^2 + a_1^2$ gives the compact expression

$$N = N_0 - 2a_1(x_w c_1 + y_w s_1). \tag{13}$$

Both N and Y are affine in (c_1, s_1). Substitution of (12) into $s_3^2 + c_3^2 = 1$ gives

$$4a_2^2(Y - d_2)^2 + (N + K - 2d_2 Y)^2 = 4a_2^2 d_4^2. \tag{14}$$

Introduce the half-angle variable $t = \tan(q_1/2)$, so that

$$D = 1 + t^2, \qquad c_1 = \frac{1 - t^2}{D}, \qquad s_1 = \frac{2t}{D}. \tag{15}$$

Writing $Y = \widetilde{Y}/D$ and $N = \widetilde{N}/D$, their quadratic numerators are

$$\begin{aligned} \widetilde{Y}(t) &= c_\gamma\Big[y_w(1 - t^2) - 2x_w t\Big] + \tfrac{1}{2}hD, \\ \widetilde{N}(t) &= N_0 D - 2a_1\Big[x_w(1 - t^2) + 2y_w t\Big]. \end{aligned} \tag{16}$$

Multiplying (14) by D^2 gives

$$\boxed{P(t) = 4a_2^2(\widetilde{Y} - d_2 D)^2 + (\widetilde{N} + KD - 2d_2\widetilde{Y})^2 - 4a_2^2 d_4^2 D^2 = 0.} \tag{17}$$

Each quantity being squared is quadratic in t, so P has degree at most four. Every real root supplies a candidate $q_1 = 2\arctan t$. The substitution $t = \tan(q_1/2)$ is undefined at $q_1 = \pi$, so this angle is not represented by any finite root of $P(t)$. To avoid missing a solution, we test $q_1 = \pi$ separately in (14) and retain it only if that equation is satisfied.

4. Recover q_2 and q_3 for each candidate q_1. For each candidate q_1, evaluate (9) and (12), then define $A = a_2 + d_4 c_3$. The remaining components of (10) are

$$X = c_2 A + s_2 d_3, \qquad Z = -s_2 A + c_2 d_3. \tag{18}$$

Thus

$$\begin{aligned} C_2 &= AX + d_3 Z = (A^2 + d_3^2)c_2, \\ S_2 &= d_3 X - AZ = (A^2 + d_3^2)s_2, \end{aligned} \qquad q_2 = \operatorname{atan2}(S_2, C_2). \tag{19}$$

The common factor is positive, so no division is needed before applying atan2. The elbow angle is recovered as $q_3 = \operatorname{atan2}(s_3, c_3)$. Since A uses c_3 directly, q_2 and q_3 can be computed in either order.

If leading polynomial coefficients vanish, the degree reduces. For this robot, $P \equiv 0$ occurs when the target wrist centre is a reachable point on the base rotation axis. At such a position, q_1 is arbitrary, yielding a continuous family of arm configurations.

Practical evaluation

Given (R_d, p_d), compute p_w from (8), followed by h, K and N_0, and form P using (16)–(17). For $P \not\equiv 0$, collect $q_1 = 2\arctan t$ from every real root and test $q_1 = \pi$ separately in (14); for $P \equiv 0$, choose any q_1, for example 0, to represent the continuous family. For

each candidate, evaluate (X, Y, Z) from (9), set $N = X^2 + Y^2 + Z^2$, obtain (s_3, c_3) from (12), and set $A = a_2 + d_4 c_3$. Then compute

$$q_2 = \operatorname{atan2}(d_3 X - AZ,\ AX + d_3 Z), \qquad q_3 = \operatorname{atan2}(s_3, c_3).$$

Check each triple against the target wrist position using forward kinematics, represent angles in $[-\pi, \pi]$ and remove duplicates modulo 2π. Pass the valid arm configurations to the wrist-orientation problem.

2.2 Endpoint orientation

For each arm configuration (q_1, q_2, q_3) found in the positioning step, we now determine the wrist angles (q_4, q_5, q_6) that give the desired orientation R_d. Different arm configurations generally require different wrist angles, so this calculation is repeated for each arm solution.

Derivation

1. Express the desired orientation relative to the arm. Using the same URDF link frames as in the positioning derivation, the orientation of `link_3_d3` relative to `base_link` is

$$R_{\text{arm}} = R_z(q_1) R_x(\gamma) R_y(q_2) R_z(q_3), \qquad \gamma = \pi/6. \tag{20}$$

The wrist joints rotate about their local x, z and x axes, respectively. There are no fixed rotations between these URDF link frames. Thus

$$R_d = R_{\text{arm}} R_{\text{wrist}}, \qquad R_{\text{wrist}} := R_{\text{arm}}^T R_d = R_x(q_4) R_z(q_5) R_x(q_6). \tag{21}$$

Here the transpose undoes the known arm rotation. Let r_{ij} denote the entry in row i and column j of R_{wrist}.

2. Use the first column to recover q_5 and q_4. Since $R_x(q_6) e_x = e_x$, the first column is

$$R_{\text{wrist}} e_x = R_x(q_4) R_z(q_5) e_x = \begin{pmatrix} c_5 & c_4 s_5 & s_4 s_5 \end{pmatrix}^T = \begin{pmatrix} r_{11} & r_{21} & r_{31} \end{pmatrix}^T. \tag{22}$$

Thus $c_5 = r_{11}$. Using $c_4^2 + s_4^2 = 1$ also gives $\rho := \sqrt{r_{21}^2 + r_{31}^2} = |s_5|$. For $\rho > 0$, take each sign $s_5 = \pm\rho$. Dividing the last two entries by s_5 gives c_4 and s_4, so the two angle pairs are

$$q_5 = \operatorname{atan2}(s_5, r_{11}), \qquad q_4 = \operatorname{atan2}\left(\frac{r_{31}}{s_5}, \frac{r_{21}}{s_5}\right). \tag{23}$$

Keep the chosen sign of s_5 in both fractions. The case $\rho = 0$ is treated in step 4.

3. Undo q_4 to recover q_6. Once q_4 is known, multiply on the left by $R_x(-q_4)$ to undo its rotation:

$$B := R_x(-q_4) R_{\text{wrist}} = R_z(q_5) R_x(q_6). \tag{24}$$

The third row of $R_z(q_5)$ is $(0, 0, 1)$, so the third row of B is $(0, s_6, c_6)$. We can therefore recover q_6 without dividing by s_5:

$$q_6 = \operatorname{atan2}(B_{32}, B_{33}). \tag{25}$$

When $\rho > 0$, the two wrist solutions are related by $(q_4, q_5, q_6) \mapsto (q_4 + \pi, -q_5, q_6 + \pi)$, with all angles taken modulo 2π.

4. Handle the cases where $s_5 = 0$. When $\rho = 0$, we have $q_5 = 0$ or $q_5 = \pi$ modulo 2π. The formula for q_4 divides by s_5 and no longer applies. The axes of joints 4 and 6 lie on the same line, so their angles cannot be determined separately. The wrist rotation reduces to $R_x(q_4 + q_6)$ for $q_5 = 0$ and $R_x(q_4 - q_6)R_z(\pi)$ for $q_5 = \pi$. Thus only a sum or a difference is fixed:

$$\begin{aligned} q_5 = 0 \ (r_{11} = 1): &\quad q_4 + q_6 = \sigma := \operatorname{atan2}(r_{32}, r_{22}), \\ q_5 = \pi \ (r_{11} = -1): &\quad q_4 - q_6 = \delta := \operatorname{atan2}(-r_{32}, -r_{22}). \end{aligned} \tag{26}$$

These relations hold modulo 2π. We may choose any q_4 and set $q_6 = \sigma - q_4$ in the first case or $q_6 = q_4 - \delta$ in the second. Choosing $q_4 = 0$ gives the representatives $(0, 0, \sigma)$ and $(0, \pi, -\delta)$ for (q_4, q_5, q_6).

Practical evaluation

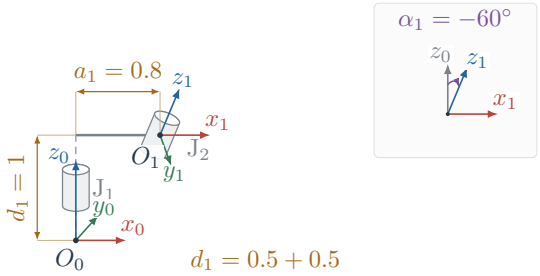
For each arm solution, use both wrist branches from steps 2–3 when ρ exceeds a small numerical tolerance. Otherwise, take $q_4 = 0$ and use the sign of r_{11} to select the case in (26). Combine each wrist solution with its arm solution, represent all angles in $[-\pi, \pi]$, and remove duplicates modulo 2π. Verify the position and orientation against (p_d, R_d) by forward kinematics, using separate tolerances.

A DH frame constructions

Figure 1 collects all six frame-to-frame constructions at the home configuration ($q = 0$), using one fixed viewing direction and length scale. Lengths are in metres; angles are shown in degrees for readability.

1 Base to shoulder: $\{0\} \to \{1\}$

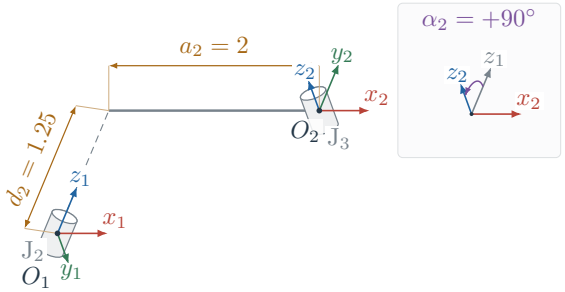
$d_1 = 1, \quad \theta_1 = q_1, \quad \alpha_1 = -60°, \quad a_1 = 0.8$



The full base height is 1 m. The 30° URDF roll of joint 2 gives $\alpha_1 = 30° - 90° = -60°$.

2 Shoulder to elbow: $\{1\} \to \{2\}$

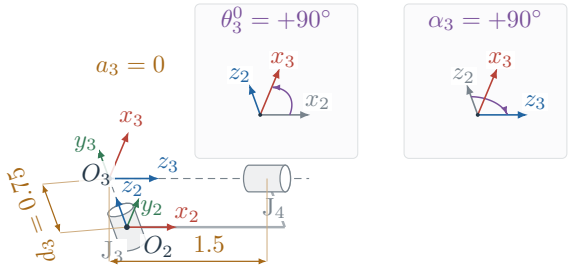
$d_2 = 1.25, \quad \theta_2 = q_2, \quad \alpha_2 = 90°, \quad a_2 = 2$



The x axes remain parallel. The offset along z_1 and the common normal along x_2 locate O_2 on joint 3.

3 Axis intersection: $\{2\} \to \{3\}$

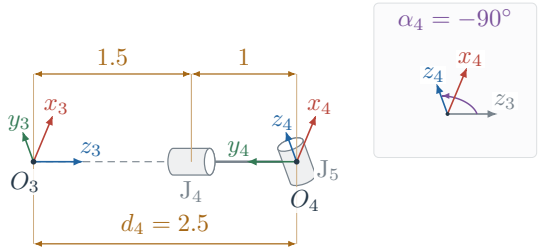
$d_3 = 0.75, \quad \theta_3 = q_3 + 90°, \quad \alpha_3 = 90°, \quad a_3 = 0$



O_3 is the intersection of the extended axes. Choosing $x_3 = y_2$ explains the fixed +90° offset in θ_3.

4 Reach the wrist: $\{3\} \to \{4\}$

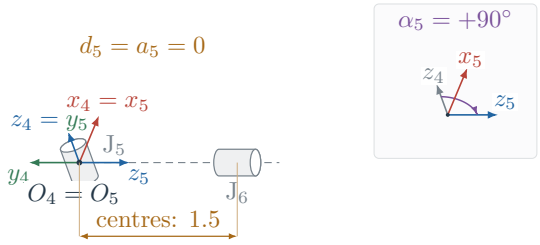
$d_4 = 2.5, \quad \theta_4 = q_4, \quad \alpha_4 = -90°, \quad a_4 = 0$



The distance starts at O_3, rather than the joint-4 cylinder. The two physical offsets add: $d_4 = 1.5 + 1$.

5 Coincident origins: $\{4\} \to \{5\}$

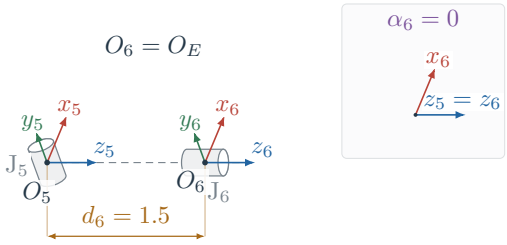
$d_5 = 0, \quad \theta_5 = q_5, \quad \alpha_5 = 90°, \quad a_5 = 0$



The origins coincide, although the joint centres do not. Here $x_4 = x_5$ and the extended joint-6 axis passes through the wrist centre.

6 Wrist to endpoint: $\{5\} \to \{6\}$

$d_6 = 1.5, \quad \theta_6 = q_6, \quad \alpha_6 = 0, \quad a_6 = 0$



At home the two frames have parallel axes. $O_6 = O_E$; the fixed endpoint alignment changes orientation only.

Figure 1: Standard DH frame assignment. Cylinders denote physical revolute joints J_i; dots mark DH origins O_i. x, y and z axes retain their colours. Dashed lines extend joint axes; gold dimensions locate the origins. Boxed angle sketches show the rotation $x_2 \to x_3$ about $+z_2$, giving θ_3^0, and the signed twist $z_{i-1} \to z_i$ about $+x_i$ (with the first direction translated to the same point). Cylinder orientation, size and viewing direction are reused between panels; drawings are schematic.